

## Multiple Choice (10 marks)

1. The \_\_\_\_\_ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
  - (a) Haunted web
  - (b) World Wide Web
  - (c) Dark web
  - (d) Surface web
2. Which of the following is not a transport layer vulnerability?
  - (a) Mishandling of undefined, poorly defined variables
  - (b) The Vulnerability that allows "fingerprinting" & other enumeration of host information
  - (c) Overloading of transport-layer mechanisms
  - (d) Unauthorized network access
3. Which of the following statements about a remote procedure call is true?
  - (a) It is used to call procedures with addresses that are farther than  $2^{16}$  bytes away.
  - (b) It cannot return a value.
  - (c) It cannot pass parameters by reference.
  - (d) It cannot call procedures implemented in a different language.
4. If  $p(x)$  is the minimal-degree interpolating polynomial for the real-valued function  $f(x)$  at the  $n + 1$  distinct real numbers  $x_0, \dots, x_n$  what is the maximum possible degree of  $p(x)$ ?
  - (a)  $n$
  - (b)  $n + 1$
  - (c)  $n + 2$
  - (d)  $2n$
5. Which of the following instruction-set features is NOT generally considered an obstacle to aggressive pipelining of an integer unit?
  - (a) Condition codes set by every instruction
  - (b) Variable-length encoding of instructions
  - (c) Instructions requiring widely varying numbers of cycles to execute
  - (d) Several different classes (sets) of registers

## True / False (10 marks)

*Answer True or False*

6. True or False: A ping sweep is used to identify live systems on a network by sending ICMP echo requests to multiple IP addresses.
7. True or False: Public key encryption is generally faster than symmetric key cryptography due to its unique algorithms.
8. True or False: The language  $\{ww \mid w \text{ in } (0 + 1)^*\}$  is context-free and can be accepted by some pushdown automaton.
9. True or False: The digest created by a hash function is commonly referred to as a Modification Detection Code (MDC).
10. True or False: In a buffer-overflow attack, cybercriminals exploit vulnerabilities by injecting malicious data to manipulate system behavior.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to calculate the harmonic sum of  $n-1$ .
12. Write a python function to find the sum of squares of binomial co-efficients.

*End of Paper*